

Research papers

Preferential flow in the vadose zone and interface dynamics: Impact of microbial exudates



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ABSTRACT

In the hydrological cycle, the infiltration process is a critical component in the distribution of water into the soil and in the groundwater system. The nonlinear dynamics of the soil infiltration process yield preferential flow which affects the water distribution in soil. Preferential flow is influenced by the interactions between water, soil, plants, and microorganisms. Although the relationship among the plant roots, their rhizodeposits and water transport in soil has been the subject of extensive study, the effect of microbial exudates has been studied in only a few cases. Here the authors investigated the influence of two artificial microbial exudates—catechol and riboflavin—on the infiltration process, particularly unstable fingered flow, one form of preferential flow. Flow experiments investigating the effects of types and concentrations of microbial exudates on unstable fingered flow were conducted in a two-dimensional tank that was filled with ASTM C778 graded silica sand. The light transmission method (LTM) which is based on capturing the light intensity transmitted through a sand-water system and then converting it into degree of water saturation was used to visualize and characterize the flow of water in porous media as well as to image and measure the spatial and temporal distribution of water in porous media. Flow patterns, vertical and horizontal profiles of the degree of water saturation of the fingers, as well as measurements of the fingers dimension (width), number, and velocity were determined using the light transmission method. Interfacial experiments exploring the influence of microbial exudates on the wettability behavior of water were performed by measuring the contact angle and the interfacial tension of the (solid)-gas-microbial exudate solution systems. Unstable wetting front generating fingered flow was observed in all infiltration experiments. The experimental results showed that the microbial exudate addition affected the infiltration process, as the measurements of the degree of saturation profiles and widths of the fingers differed from those of the control NaCl solution. These differences may be due to an improved water holding capacity in the presence of the microbial exudates. The lowest catechol solution concentration (10 μM) produced the largest finger width (9.69 cm) among the tested catechol solution concentrations and all the other solutions including the control solution (7.24 cm). Moreover, the wettability of the medium for the catechol solution increased with an increase in concentration. The highest riboflavin solution concentration (1000 μM) generated the highest finger width (7.75 cm) among the tested riboflavin solution concentrations. However, the wettability of the medium for the riboflavin solution decreased with an increase in concentration. Our study demonstrated that the microbial exudates which are biochemical compounds produced and released by microbes in the environment are capable of influencing the soil infiltration process. The results of this study also demonstrated that the influence of the contact angle expressed as $(\cos \theta)^{1/2}$ should be integrated in the scaling of the finger dimension, i.e., finger width, when the Miller and Miller (1956) scaling theory is applied for the hydrodynamic scaling in porous media.

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1. Introduction

The infiltration process is an extremely important factor within the hydrological cycle as it filters water from both the atmosphere and the ground surface into the soil, which in turn revives the underground ecosystem through a recharged ground water table to ensure the continuity of the cycle. A combination of gravity and capillarity forces acts upon the water and upon entry into the soil pores with gravity always acting vertically and with the capillary forces channeling the water in all directions (Gray and Norum, 1967). Derived in 1931 by Lorenzo A. Richards, the Richards equation is used to determine water flow in the vadose zone under constant temperature from Darcy's law and the conservation of mass. It is expressed as:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(\theta) \frac{d\psi(\theta)}{d\theta} \frac{\partial \theta}{\partial z} \right] - \frac{\partial K(\theta)}{\partial z} = \frac{\partial}{\partial z} \left[D(\theta) \frac{\partial \theta}{\partial z} \right] - \frac{\partial K(\theta)}{\partial z} \quad (1)$$

where θ represents the volumetric water content in soil, t the time, z the vertical axis pointing downward indicating positive, K the soil hydraulic conductivity, ψ the soil water potential, and D the soil water diffusivity. This highly nonlinear problem makes analytical and numerical solutions to the Richards equation typically non-unique (Celia et al., 1990). The theory of infiltration and solutions to elucidate the question of flow in porous media have been the focus of extensive research, most particularly through the seminal work of Jean-Yves Parlange (Parlange, 1971a,b,c, 1972a,b,c,d, e, 1973, 1975; Parlange and Aylor, 1972). Analytical and quasi-analytical solutions of the equation of flow in porous media have been formulated through time expansion (Philip, 1969, 1975a,b), integral approaches (Parlange, 1971b, 1972e) and the nonmonotonic traveling wave (DiCarlo et al., 2008). Many simplifications and approximations to the Richards equation have been proposed (e.g. Bras, 1990; Smith et al., 1993; Waechter and Philip, 1985).

Water flow during the infiltration process in soil systems is either uniform, with wetting fronts flowing in parallel to the soil surface level; or non-uniform, with wetting fronts flowing irregularly (Green and Ampt, 1911). In a non-uniform flow, water movement in some areas of the unsaturated subsurface is faster and more intense than in others. Preferential flow in the form of macropore flow, unstable fingered flow, and funnel flow are a few of the characteristic non-uniform flows that may occur. Preferential flow refers to the movement of water in addition to the solutes in a non-uniform flow and is independent from common conditions that characterize soils in various locations (Andreini and Steenhuis, 1990; Baveye et al., 1998; Dekker and Ritsema, 1996; Doerr et al., 2007, 2006). The occurrence of finger flows, one form of preferential flow that frequently occurs in homogeneous porous media, may appear due to (i) entrapment of air during the movement of the wetting front, (ii) flow through water-repellent soils, (iii) the process of continuous non-ponding infiltration, and (iv) flow through layered materials having different textures (Baker and Hillel, 1990; Diment and Watson, 1985; Du et al., 2001; Glass et al., 1991, 1990, 1989a,b,c; Hill and Parlange, 1972; Hill and Baker, 1988; Parlange and Hill, 1976; Philip, 1975a,b; Raats, 1973; Selker et al., 1992; Tamai et al., 1987; Yao and Hendrickx, 1996, 2001). Surface tension and contact angle are the two interfacial parameters most commonly examined in the infiltration process (Yuan and Lee, 2013). Additional details on wetting front instability and gravity-driven flow in porous media can be obtained in extensive reviews (Assouline, 2013; Glass and Nicholl, 1996; De Rooij, 2000; DiCarlo, 2013; Xiong, 2014).

The formation of preferential flow may be influenced either directly or indirectly from various biological factors. Although plant root zones induce water into the soil, microorganisms and their secreted compounds can change hydraulic conductivity by

modifying soil porosity and inducing water repellency, which then forms preferential flow paths (Morales et al., 2010). These preferential flow paths of a soil profile can then form a saturated distribution zone with fingering flow paths (Kim et al., 2005; Steenhuis et al., 1994). These unstable wetting fronts then split the original horizontal profiles into fingering flow paths that may cause a leaching of contaminants via the soil into the groundwater (Bauters et al., 2000; Darnault et al., 2003, 2004; Doerr et al., 2007; Hendrickx and Flury, 2001; Uyusur et al., 2010, 2015; Wang et al., 2000). In addition, unexpected dry and wet soil patches from fingering flow paths can negatively affect microbial populations and plant growth (Morales et al., 2010). For example, the agrochemicals and wetting agents (surfactants) regularly applied to golf course greens can rapidly move through the subsurface by preferential flow (Kostka, 2000; Morales et al., 2010).

Soil moisture is an integral component concerning the support of biogeochemical processes as nutrient cycles, and a catalyst for microbial and plant activities (Wang et al., 2015). Soil modification by microorganisms and plant roots creates preferential flow, which in turn facilitates the microbial and plant activity by supplying more oxygen, moisture content, available nutrients, and different dissolved substrates along the wetted flow path. The soil-water dynamics are complex due to the heterogeneity in soil media from the nonuniform spatial distribution of microorganisms and plant roots, different soil structures, numerous interrelated biogeochemical reactions, spatial and temporal variability, and the scaling factor for hydrological processes (Ettema and Wardle, 2002; Morales et al., 2010).

Microorganisms growing in the soil tend to accumulate low molecular weight organic molecules, also known as osmolytes. The semi-permeability of membranes of these microorganisms causes a release of the intracellular solutes as exudates, which in turn preserve their osmotic potential with the surrounding medium. Although the exudate production varies among microorganisms, this microbial response is a useful strategy to prevent cell lysis and death caused by changes in soil water content (Boot et al., 2013; Griffin, 1981; Halverson et al., 2000; Kieft et al., 1987; Potts, 1994; Wang et al., 2015). It has been determined that plant exudates released by roots can function as surfactants and change surface tension of soil solutions, which in turn alters the hydraulic conductivity and water flow around the rhizosphere (Passioura, 1988; Read et al., 2003; Read and Gregory, 1997). As surface-active organic compounds, surfactants can alter both surface tension of liquids and contact angle at the solid-liquid interface, which can further influence both the water flow and the potential transport of contaminants in the vadose zone (Bashir et al., 2008; Karagunduz et al., 2015). However, little research has been undertaken to examine the influence of different microbial exudates on water infiltration processes into the vadose zone. Catechol is one compound produced by certain bacteria (e.g., *Escherichia coli*) that can be synthesized by feeding those microbes on glucose and as intermediates during the biodegradation processes of aromatic compounds (e.g., phenol) (Draths and Frost, 1995; Li et al., 2005; Nair et al., 2008). Riboflavin (or vitamin B₂), which is naturally produced by many microorganisms, including *Candida famata*, *Bacillus subtilis*, and lactic acid bacteria, is a second compound that can be treated as a microbial exudate (Bacher et al., 2001; Perkins et al., 1999; Perkins and Pero, 1993; Schallmeyer et al., 2004; Stahmann et al., 2000; Thakur et al., 2016).

First used by Hoa in 1981 to measure water content in two-dimensional unsaturated sandy porous media, the Light Transmission Method (LTM) is ideal for studying the infiltration process without disturbing the porous media system. This non-invasive imaging technique is inexpensive and fast responding, and provides high resolution with a minimum of acquisition time (Darnault et al., 1998; Werth et al., 2010). In order to acquire

available data, a transparent (thin enough to be two dimensional) system that enables visible light to transmit through is needed. A charge-coupled device (CCD) camera is commonly used for data capture in the light transmission method, as the resolution is acceptable (Werth et al., 2010). The light intensity data separated from the original RGB image can be converted to the percentage of water saturation based on the calibration equation. As a result, profiles of water distribution of wetting front and fingered flow during the infiltration process can be obtained and then analyzed for flow characteristics.

Very little study has been undertaken to examine the infiltration process in porous media using the light transmission method, and even less to elucidate the influence of microbial exudate on the flow process. Although plant exudates act as surfactants to influence water flow, the effect of microbial exudates upon the infiltration process is unknown. Therefore, a random rain event, rather than point source, was simulated to mimic natural conditions, and to use the light transmission method to investigate the effect of these microbial exudates on preferential flow with different concentrations which relate to different concentrations of biomass when applying real microbes.

The primary objective of this research entailed an investigation of how different microbial exudates with varying concentrations alter the infiltration process in an unsaturated sand system and the distribution of the soil moisture content. Specifically, the microbial exudates catechol and riboflavin were examined to determine any influence on flow in porous media. The surface tension and the contact angle were dynamically monitored to obtain the interfacial properties of exudate solutions, and the infiltration experiments were conducted in a homogeneous and initially dry porous media in a two-dimensional (2D) tank. A light transmission method with high spatial and temporal resolution was used to construct the flow patterns and to visualize the water saturation distribution followed by the use of MATLAB to process the imaging and data analyses.

2. Material and methods

2.1. Microbial exudates solution preparation

Two solutions of microbial exudate were simulated using catechol and riboflavin, the molar concentrations of which were 10, 100, 500, and 1000 μM , respectively. For the catechol ($\geq 99.5\%$, Sigma-Aldrich, St. Louis, MO) solution, 1.10, 11.01, 55.05, and 110.10 mg of catechol powder were mixed with 0.01 M of BDH ACS Grade NaCl (VWR International, Radnor, PA) solution. For the riboflavin ($\geq 98\%$, Sigma-Aldrich, St. Louis, MO) solution, 3.76, 37.64, 188.18, and 376.36 mg of riboflavin powder were mixed with 0.01 M of NaCl solution. Ultrapure deionized (DI) water with a resistivity of $18.2 \text{ M}\Omega\cdot\text{cm}$ (Millipore Corporation, Billerica, MA) was used to limit any errors from impurities. A solution of 0.01 M NaCl was used as the control. Prior to measurements, all solutions were adjusted to pH of 4.80 ± 0.03 by adding 0.01 M of HCl (36.5–38.0%, Sigma-Aldrich, St. Louis, MO) or 0.01 M of BDH ACS Grade NaOH (VWR International, Radnor, PA) using a XL 500 pH meter (Fisher scientific, Waltham, MA). The different constituents and concentrations of the solutions are listed in Table 1.

2.2. Contact angle and surface tension

A Drop Shape Analyzer, the FM40Mk2 EasyDrop model instrument (Krüss, Hamburg, Germany), was used to measure the contact angle of the different experimental solutions, and the EasyDrop complimentary Drop Shape Analysis software was used to process the data. A 5 μL drop of sample was excreted from a 1 mL Norm-Ject syringe (Fisher Scientific, Waltham, MA) with an

attached needle to form a sessile drop on a glass slide that was continuously measured at 30 s increments when placed in front of a camera for 10 min. All samples were conducted in triplicate, and the results were recorded in Excel.

A Krüss Drop Shape Analyzer apparatus with a needle of approximately 1.8 mm in diameter was then used to experimentally measure the surface tension of the different microbial exudate solutions and the control solution. A pendant drop was formed from the tip of the needle and imaged by the camera to determine the interfacial tension measurement in mN/m . The parameters for embedding phase density (air in this case) and temperature were adjusted in the setting to 0.0012 g/cm^3 and 23°C for the surface tension measurements. Droplets from the sample solutions were measured at 30 s increments in front of the camera over five full minutes. All samples were conducted in triplicate, and the results were recorded in Excel. An overview of the experimental parameters and analyses for the interfacial dynamics study is outlined in Table 1.

2.3. Two-dimensional tank flow experiment

A two-dimensional tank with two plates constructed of scratch-resistant polycarbonate and an inner dimension of $30.2 \text{ cm} \times 1.2 \text{ cm} \times 30.8 \text{ cm}$ ($\text{L} \times \text{W} \times \text{H}$) was used for the microbial exudate flow experiments. The bottom of the tank on which was layered four pieces of ceramic porous plates for permeability was inclined downward to the left. Four evenly distributed drainage outlets and one additional outlet at the left corner were also placed within the tank. During flow experiments, a filter sheet was placed at the bottom to prevent sand clogging and for the proper solutions to drain. A picture of the 2D tank with setup is shown in Fig. 1. ASTM C778 Graded Sand with unground silica (U.S. Silica Company, Ottawa, IL) was used to fill the tank. The physical and chemical characterization of the ASTM C778 Graded sand from the US Silica product data sheet are provided in Table 2. A sand volume corresponding to the volume of two tanks was loaded into a handmade Y-shaped sand loader with a piece of cloth within, with the loader taped above the tank. The cloth was then quickly removed from the tank to create a uniform packing of sand within the tank with a uniform sand bulk density of 1.50 g/cm^3 . The rest of the sand in the loader and the loader were then removed. To ensure a consistent density of sand, a plastic rod was used to tap the top edge of the tank several times to “settle” the top sand layer.

After the tank was loaded with sand, the microbial exudate solutions and the control solution were pumped into the tank through a Model 77200-60 Easy-Load II peristaltic pump head (Masterflex, Gelsenkirchen, Germany) via a connecting tube at a constant flow rate of 10 mL/min. Solutions were continuously stirred in their containers which were positioned on a MS3002S NewClassic balance (Mettler Toledo, Columbus, OH) to record the initial and final weights. The other end of the tube was connected to an actuator, which moved back and forth at a constant speed above the tank. Two short tube openings were suspended from the actuator, and solutions were randomly dropped from either tube to simulate a random rainfall event with an infiltration velocity of 13.12 cm/hr into the sand system. An overview of the experimental parameters and analyses for the two-dimensional tank flow in porous media study is outlined in Table 1.

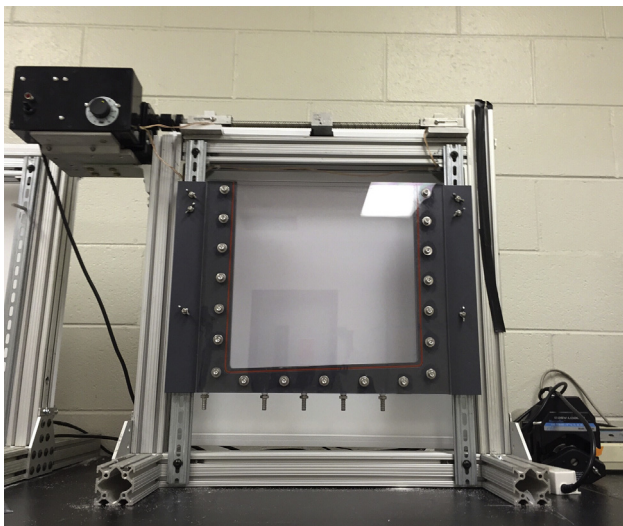
2.4. Visualization and light transmission method

The light transmission method was used to visualize the microbial exudate solution flow pattern within the 2D tank and to measure the spatial and temporal water saturation in porous media. A $0.6 \text{ m} \times 0.6 \text{ m}$ square even-glow LED panel light fixture with natural white color and a maximum of 2900 lm (Super Bright LEDs,

Table 1

Overview of the experimental parameters and analyses to characterize the interfacial dynamics and flow in porous media.

Experiments and analyses	Solutions								
	Composition								
	NaCl Concentration (M)	NaCl (0.01 M) + Catechol Catechol concentration (μ M)				NaCl (0.01 M) + Riboflavin Riboflavin concentration (μ M)			
	0.01	10	100	500	1000	10	100	500	1000
Interfacial dynamics									
Surface tension measurement									
Number of replicates	3	3	3	3	3	3	3	3	3
Frequency of measurement (s)	30	30	30	30	30	30	30	30	30
Duration of measurement (min)	5	5	5	5	5	5	5	5	5
Contact angle measurement									
Number of replicates	3	3	3	3	3	3	3	3	3
Frequency of measurement (s)	30	30	30	30	30	30	30	30	30
Duration of measurement (min)	10	10	10	10	10	10	10	10	10
Flow in porous media									
Infiltration process characteristics									
Number of replicates	2	2	2	2	2	2	2	2	2
Pump flow rate ($\text{mL}\cdot\text{min}^{-1}$)	10	10	10	10	10	10	10	10	10
Flow velocity ($\text{cm}\cdot\text{min}^{-1}$)	0.22	0.22	0.22	0.22	0.22	0.22	0.22	0.22	0.22
Infiltration duration for replicates (min)	17	13–17	15–16	14–16	13–16	13–16	13–14	15–16	17–19
Data and imaging analysis of infiltration process									
Digital image acquisition frequency (s)	30	30	30	30	30	30	30	30	30
Duration of measurement (min)	~30	~30	~30	~30	~30	~30	~30	~30	~30
Water saturation degree measurement in fingered flow									
Vertical profiles in growing fingers	✓	✓	✓	✓	✓	✓	✓	✓	✓
Horizontal profile at fingertip of fully formed fingers	✓	✓	✓	✓	✓	✓	✓	✓	✓
Dimension measurement of fingered flow									
Finger width (cm)	✓	✓	✓	✓	✓	✓	✓	✓	✓
Finger length (cm)	✓	✓	✓	✓	✓	✓	✓	✓	✓
Porous media type									
US Silica Sand ASTM C778 Standard	✓	✓	✓	✓	✓	✓	✓	✓	✓

**Fig. 1.** An overall structure and configuration of the 2D tank and imaging system.

St. Louis, MO) was placed behind the 2D tank. The entire configuration except for the tank front surface was fully covered by a black plastic sheet to avoid extra light emanation from the side. A D5500 DSLR camera (Nikon, Tokyo, Japan) on a VTSL7200 72" tripod (Bower, Long Island City, NY) was placed in front of the tank at a distance to record the transmitted light through the sand column in the 2D tank with a constant flow of exudate solution. The

camera setting was adjusted to "manual" to avoid overexposure, with the LED panel light only activated when both pump and actuator started to function. The camera then began imaging automatically every 30 s until the bottom of the tank was saturated and then drained for at least five minutes. Each flow experiment was performed in duplicate. The secondary experimental images of each flow experiment are available for viewing in the [Supplementary Material \(Figures S1–S4\)](#).

An empirical calibration was conducted in calibration cell tanks ($7.6 \times 1.5 \times 8.9$ cm) packed with sand to a full range of known water saturation percentage values (i.e. 0%, 5%, 10%, 20%, 40%, 60%, 80%, and 100%). The calibration was performed in triplicate. A linear calibration relationship was observed between water saturation and intensity of the transmitted light, with an R^2 value of 0.96 (Fig. 2); this relationship was also detected by [Uyusur et al. \(2015\)](#). This linear relationship and resulting equation were then used to transform the intensity values into water saturation values with MATLAB (R2016a, The Mathworks Inc., Natick, MA, USA). Therefore, for each flow experiment, the intensity values corresponding to 0% and 100% water saturation were acquired and the slope from each experiment was evaluated to the calibration curve to confirm that the empirical calibrations among experiments were similar.

2.5. Data and imaging analysis

For each flow experiment, four pictures representing different stages of the main finger formation, prior to reaching the tank's

Table 2

Physical and chemical properties of ASTM C778 Graded Sand from US Silica product data sheet.

Physical properties		
	USA STD sieve size (mm)	Percentage sand grain retained on sieve (Individual%)
	1.180	0.0
	0.600	2.0
	0.425	28.0
	0.300	45.0
	0.150	23.0
	Pan	2.0
	Color	White
	Grain shape	Round
	Hardness (Mohs)	7
	Melting point (Degree F)	3100
	Mineral	Quartz
	Particle density (g/cm ³)	2.65
Chemical properties		
	pH	7
	SiO ₂ (%)	99.7
	FeO ₃ (%)	0.02
	Al ₂ O ₃ (%)	0.06
	TiO ₂ (%)	0.01
	CaO (%)	<0.01
	MgO (%)	<0.01
	Na ₂ O (%)	<0.01
	K ₂ O (%)	<0.01
	Loss on ignition	0.1

bottom, were analyzed, in addition to the original image at a time of 0 min and a drainage image for calibration purposes. MATLAB was used to process the images, specifically to convert RGB (red, green and blue) files to the form in HSV (hue, saturation and intensity), where solely the intensity was used later. The imaging method yielded both a high spatial resolution by tracking the main finger formation and a constant temporal response at 30-s intervals. Both vertical and horizontal water distribution profiles were

analyzed along and across the main finger for each stage of development. A 30 pixel-wide rectangle appearing as a vertical line was drawn through the middle of the main finger to provide a vertical water distribution profile along the length, with a second rectangle horizontally drawn across the main finger at 90% of its total length to provide a horizontal water distribution profile. Regression equations were calculated using the observed linear relationship between the degree of water saturation and the image of transmitted light intensity (Fig. 2) based on the initial dry sand and fully saturated sand images with water saturation degree of 0% and 100%, respectively (Darnault et al., 2001; Hoa, 1981; Uyuşur et al., 2016). MATLAB was then used to code the regression equations to calculate the water saturation based on the transmitted light intensity values of each finger. An overview of the imaging parameters and analyses for the two-dimensional tank flow in the porous media study is outlined in Table 1.

2.6. Statistical analysis

Interfacial parameters and flow characteristics were examined statistically to make comparisons between types and concentrations of microbial exudate solutions. ANOVA was utilized to determine the statistical significance of the experimental factors. ANOVA Student's t tests with an alpha value of 0.05 were run to determine statistically significant differences among factor means. Microsoft Excel 2010 and JMP Pro 12 were used for the statistical analysis.

3. Results

3.1. Contact angle and surface tension

3.1.1. Catechol

The contact angles for the catechol solutions are shown in Fig. 3. The lowest concentration of 10 µM catechol solution with 0.01 M NaCl exhibited the highest contact angle among the four concentrations. An increase in the catechol concentration from 10 µM to

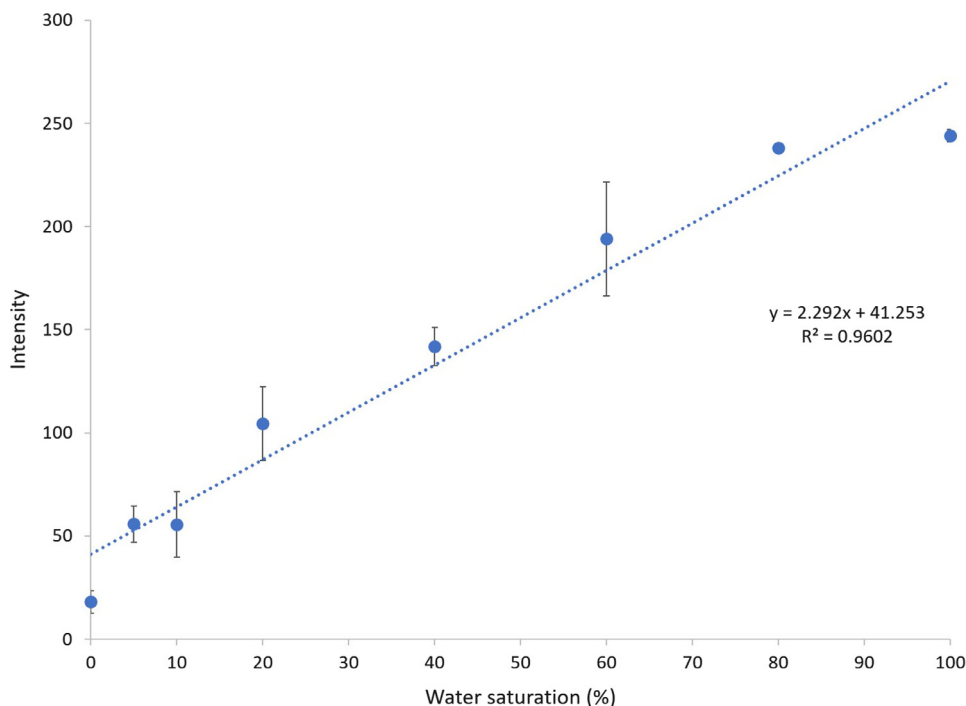


Fig. 2. Empirical calibration curve of the transmitted intensity vs. the degree of water saturation, with its linear trend line and slope equation.

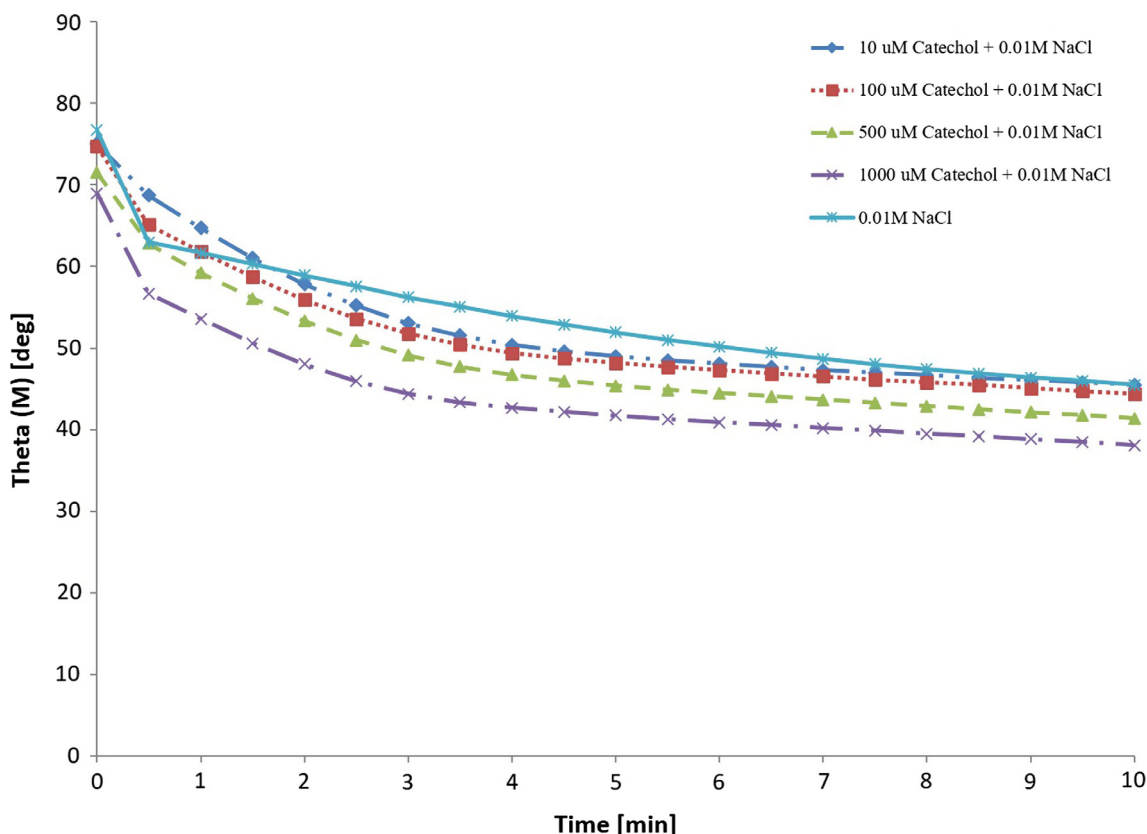


Fig. 3. Contact angle measurements for the catechol solution at concentrations of 10, 100, 500, and 1000 μM .

1000 μM reduced the contact angle from $75 \pm 2.82^\circ$ to $69 \pm 3.52^\circ$ at time 0 min successively. At approximately 10 min, all contact angles decreased and slowly plateaued. The initial contact angle of $76.7 \pm 1.61^\circ$ in the 0.01 M NaCl control solution suddenly declined and finally exhibited a higher contact angle compared to the solutions with catechol added.

The measurements of surface tension for catechol solutions in Fig. 4 show similar results between 72.5 mN/m to 74.5 mN/m within a period of 5 min. The initial surface tension for a 10 μM solution of catechol was greater than the 100 μM and 1000 μM solutions of catechol, with the lowest surface tension within the 500 μM solution. However, within five minutes, a slight fluctuation of surface tensions indicated no significant difference overall. Compared to the control solution of 0.01 M NaCl with the surface tension at 73.7 ± 0.04 mN/m, the 10 μM and 100 μM catechol solutions caused an increase in surface tension, whereas the 500 μM and 1000 μM solutions caused a decrease.

3.1.2. Riboflavin

The contact angles for the riboflavin solutions are shown in Fig. 5. Of the four concentrations analyzed, the lowest contact angle was for the 10 μM concentration of riboflavin solution with 0.01 M NaCl at 0 min, and the solution with the 1000 μM concentration exhibited the highest value. The contact angle increased from $75.3 \pm 0.70^\circ$ to $80.3 \pm 1.47^\circ$ when the riboflavin concentration increased from 10 μM to 1000 μM , the rise of which was concentration-dependent and increased in order; however, when the time approached 10 min, all decreasing trends began to plateau. Although the initial contact angle of the 0.01 M NaCl control solution was greater than 10 μM and 100 μM solutions, the overall trend was similar to the 10 μM of riboflavin solution in the later stage.

The surface tensions for riboflavin solutions were then measured continuously every 30 s for 5 min, which are shown in Fig. 6, indicating a lower surface tension with higher riboflavin concentrations. These values fluctuated slightly, however, which failed to indicate any great differences within the range of 73.5 mN/m to 75 mN/m. The initial surface tension decreased from 74.6 ± 0.12 mN/m to 73.9 ± 0.25 mN/m when the riboflavin concentration increased from 10 μM to 1000 μM . The 0.01 M NaCl control solution exhibited the lowest surface tension during this five-minute interval.

3.1.3. Microbial exudate solutions versus control

The percentage difference in contact angle for catechol and riboflavin solutions compared to the control NaCl solution is shown in Fig. 7. At concentrations of 500 μM and 1000 μM , the two microbial exudates exhibited the opposite percentages compared to the baseline of control. For example, the contact angle for catechol solution was about 11% smaller than the control, while it was almost 13% greater than the control for riboflavin solution at 1000 μM .

Similarly, the percentage difference in surface tension for catechol and riboflavin solutions compared to the control is plotted in Fig. 8. At concentrations of 500 μM and 1000 μM , the two microbial exudates also showed opposite values, although the overall difference in percentage was not large compared to that in the contact angle. For example, at 500 μM , the catechol solution was -0.36% and the riboflavin solution was 0.68% different from the control.

Diverse significant differences were found in contact angle and surface tension values both between types of microbial exudates and among concentrations, except for the first minute of contact angle measurement (Table 3).

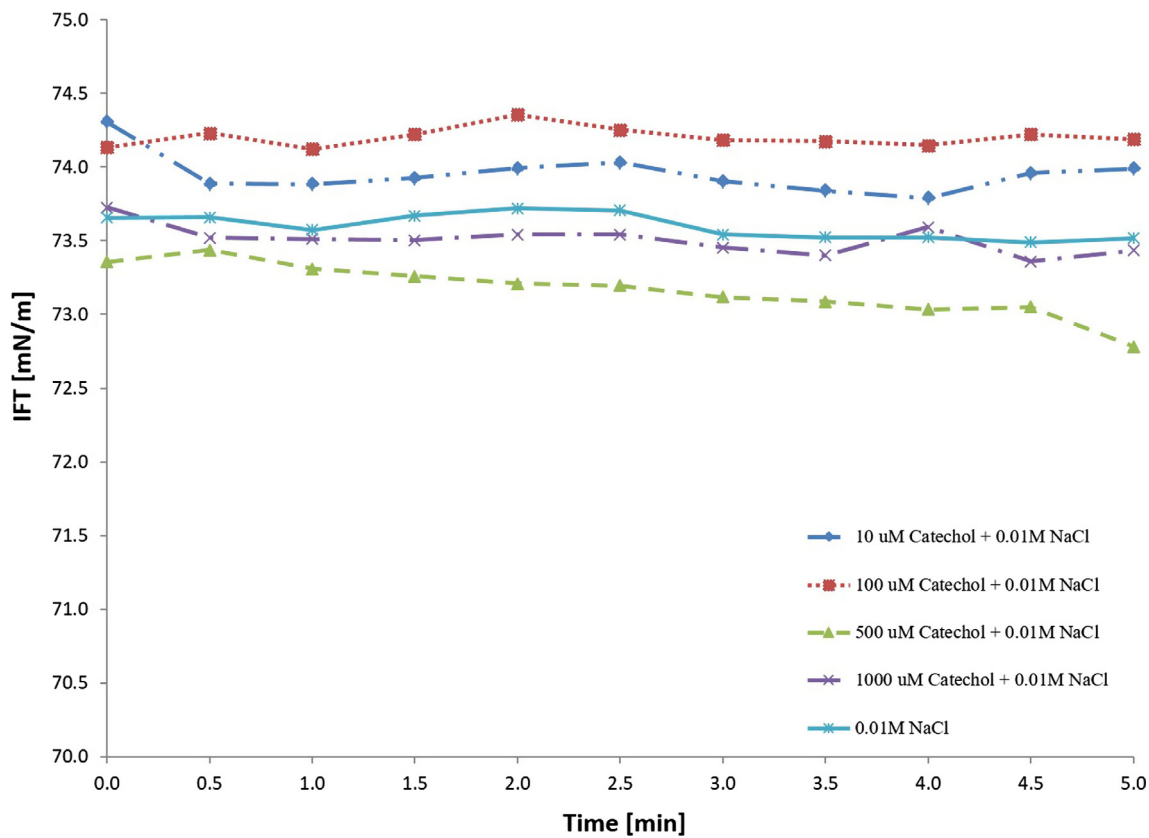


Fig. 4. Surface tension measurements for the catechol solution at concentrations of 10, 100, 500, and 1000 μM .

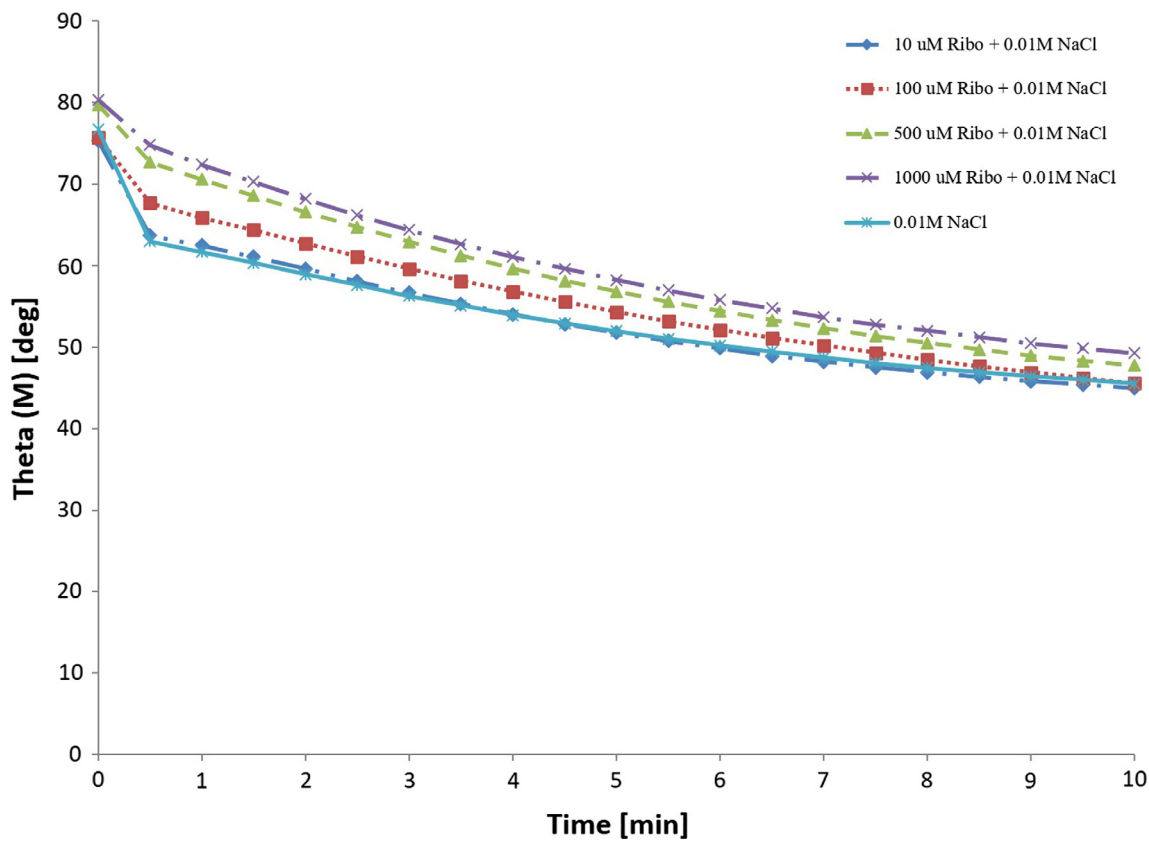


Fig. 5. Contact angle measurements for the riboflavin solution at concentrations of 10, 100, 500, and 1000 μM .

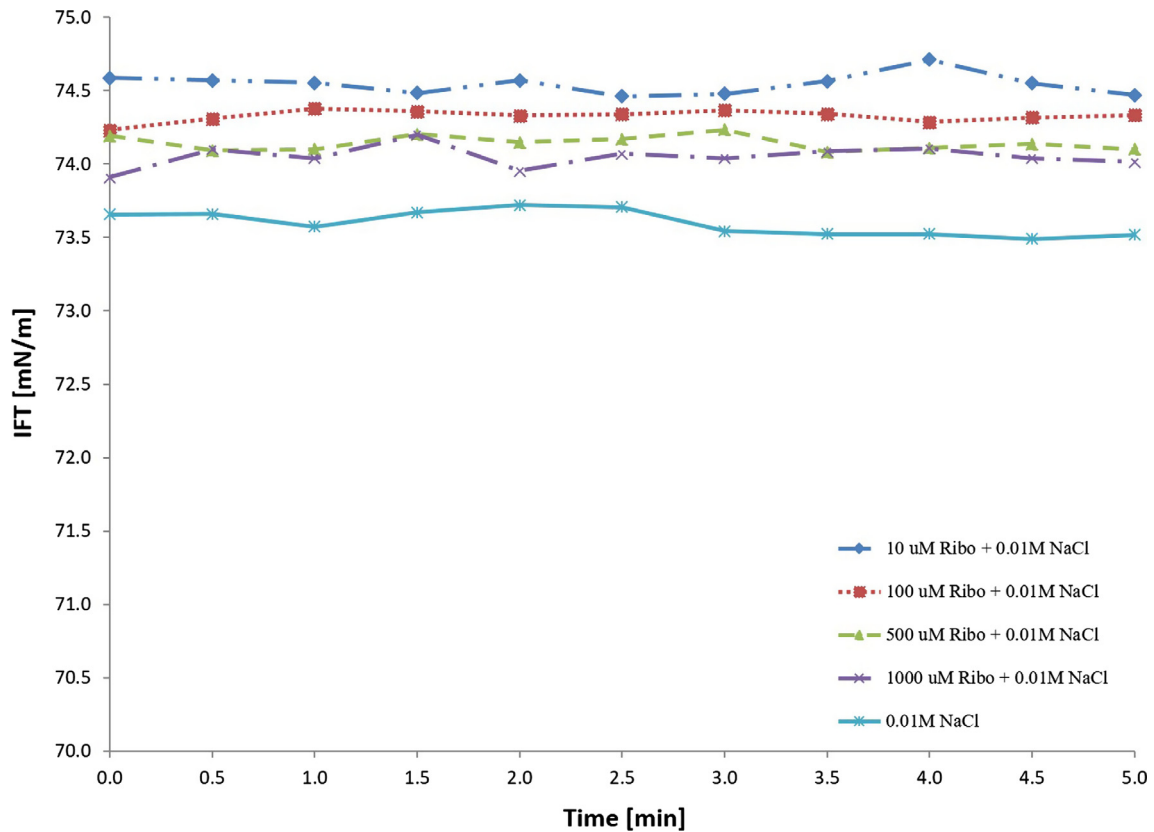


Fig. 6. Surface tension measurements for the riboflavin solution at concentrations of 10, 100, 500, and 1000 µM.

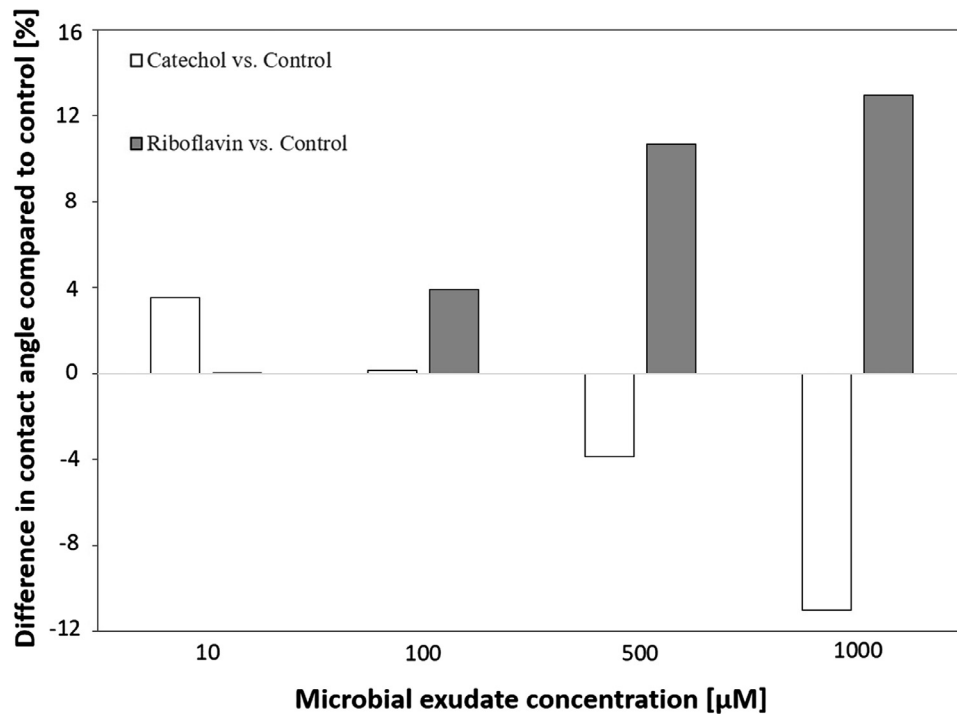


Fig. 7. Percentage change of contact angle for microbial exudate solutions compared to NaCl control.

3.2. Profile analysis of the finger development

The 2D tank visualizations of the flow experiments representing the microbial exudate dependent infiltration processes were sub-

sequently displayed for water distribution profiles as original RGB images, extracted intensity figures, and water saturation percentage profiles. Detailed information about the finger numbers and characteristics are shown in Table 4. The results from

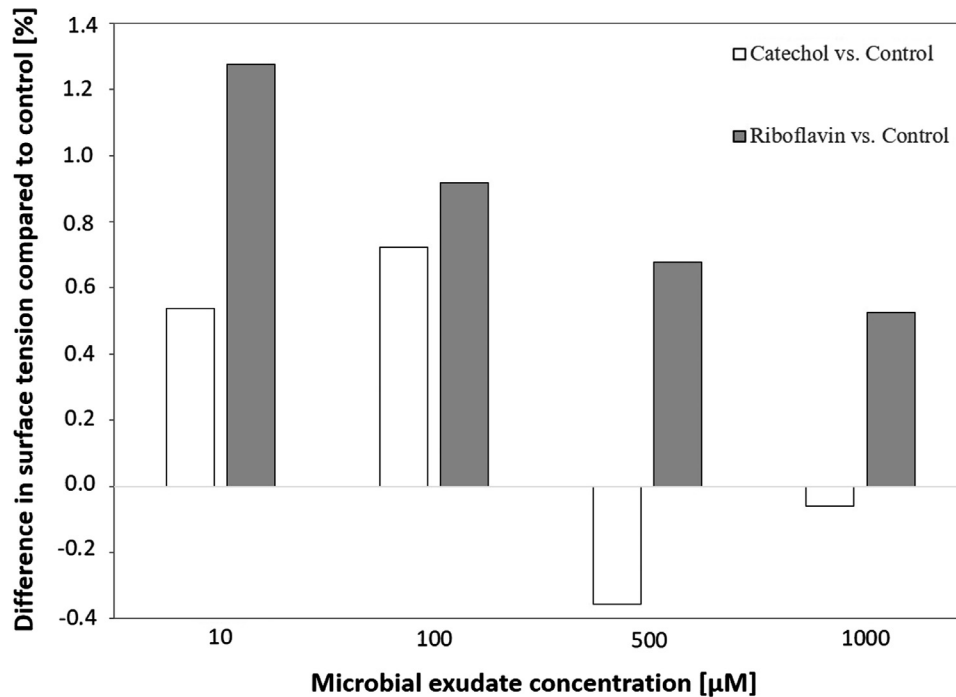


Fig. 8. Percentage change of surface tension for microbial exudate solutions compared to NaCl control.

Table 3

Characteristics of the surface tension and contact angle for catechol and riboflavin solutions and the NaCl control solution with statistical analysis of interfacial parameters with significant differences among microbial exudate solutions. Note: Values not connected by same letter are significantly different.

Solution and concentration		Contact angle (degree)		Surface tension (mN·m ⁻¹)	
		At 0 min.	For 1st min.	At 0 min.	For 1st min.
Control NaCl	0.1 M	76.70 ^{BC}	67.11 ^{ABC}	73.66 ^{DE}	73.63 ^D
Catechol	10 μM	75.00 ^C	69.48 ^{ABC}	74.31 ^{AB}	74.03 ^C
	100 μM	74.70 ^{CD}	67.22 ^{ABC}	74.13 ^{BC}	74.16 ^{BC}
	500 μM	71.53 ^{DE}	64.51 ^{BC}	73.36 ^E	73.37 ^E
	1000 μM	68.97 ^E	59.73 ^C	73.73 ^D	73.59 ^D
	1000 μM	68.97 ^E	59.73 ^C	73.73 ^D	73.59 ^D
Riboflavin	10 μM	75.30 ^C	67.13 ^{ABC}	74.59 ^A	74.57 ^A
	100 μM	75.70 ^C	69.73 ^{ABC}	74.23 ^{BC}	74.31 ^B
	500 μM	79.70 ^{AB}	74.29 ^{AB}	74.19 ^{BC}	74.13 ^{BC}
	1000 μM	80.30 ^A	75.80 ^A	73.91 ^{CD}	74.02 ^C
	1000 μM	80.30 ^A	75.80 ^A	73.91 ^{CD}	74.02 ^C

Table 4

Characteristics of the main fingers for catechol and riboflavin solutions and the NaCl control solution with statistical analysis of fingered flow parameters with significant differences among microbial exudate solutions. Note: Values not connected by same letter are significantly different.

Solution and concentration		Number of fingers			Velocity of the main finger (cm/min)			Length of main finger (cm)			Width of main finger (cm)		
		R1	R2	Avg.	R1	R2	Avg.	R1	R2	Avg.	R1	R2	Avg.
Control NaCl	0.1 M	3	2	2.5 ± 0.71 ^A	1.62	1.60	1.61 ± 0.01 ^A	27.56	27.29	27.43 ± 0.19 ^{AB}	8.40	6.08	7.24 ± 1.64 ^{AB}
Catechol	10 μM	2	1	1.5 ± 0.71 ^A	1.63	2.05	1.84 ± 0.30 ^A	27.69	26.59	27.14 ± 0.78 ^{AB}	11.72	7.66	9.69 ± 2.87 ^A
	100 μM	2	2	2.0 ± 0.00 ^A	1.76	1.82	1.79 ± 0.04 ^A	28.23	27.23	27.73 ± 0.71 ^A	6.11	7.93	7.02 ± 1.29 ^{AB}
	500 μM	2	2	2.0 ± 0.00 ^A	2.04	1.70	1.87 ± 0.24 ^A	28.54	27.21	27.88 ± 0.94 ^A	7.24	7.59	7.41 ± 0.25 ^{AB}
	1000 μM	2	2	2.0 ± 0.00 ^A	2.17	1.69	1.93 ± 0.34 ^A	28.18	26.98	27.58 ± 0.85 ^{AB}	5.08	5.16	5.12 ± 0.06 ^B
	1000 μM	2	2	2.0 ± 0.00 ^A	2.07	1.47	1.77 ± 0.42 ^A	26.94	22.87	24.91 ± 2.88 ^B	6.66	7.25	6.96 ± 0.42 ^{AB}
Riboflavin	10 μM	2	2	2.0 ± 0.00 ^A	1.87	2.18	2.03 ± 0.22 ^A	26.25	28.29	27.27 ± 1.44 ^{AB}	6.97	7.71	7.34 ± 0.52 ^{AB}
	100 μM	2	3	2.5 ± 0.71 ^A	1.88	1.74	1.81 ± 0.10 ^A	28.24	27.92	28.08 ± 0.23 ^A	9.84	4.88	7.36 ± 3.51 ^{AB}
	500 μM	2	3	2.5 ± 0.71 ^A	1.88	1.74	1.81 ± 0.10 ^A	28.24	27.92	28.08 ± 0.23 ^A	9.84	4.88	7.36 ± 3.51 ^{AB}
	1000 μM	2	2	2.0 ± 0.00 ^A	1.63	1.48	1.55 ± 0.11 ^A	27.71	28.09	27.90 ± 0.27 ^A	7.16	8.33	7.75 ± 0.83 ^{AB}
	1000 μM	2	2	2.0 ± 0.00 ^A	1.63	1.48	1.55 ± 0.11 ^A	27.71	28.09	27.90 ± 0.27 ^A	7.16	8.33	7.75 ± 0.83 ^{AB}

statistical analysis showed that there was no significant difference within the number of fingers and the velocity of finger formation among types and concentrations of solutions (Table 4). In addition, main finger length and width did not significantly differ from each other.

3.2.1. Catechol

Vertical and horizontal finger profiles were analyzed for different concentrations of catechol solutions at different stages of the main finger formation. These four successive stages, beginning at 5 min and ending prior to contact of the primary finger to the tank

bottom are shown in their vertical water saturation profiles and respective catechol concentrations (Fig. 9(a) to (d)). It was determined, from a comparison of the main finger shapes between catechol concentrations that more small branches formed on the two sides of the finger at higher concentrations. The finger exhibiting the most tendrils was that observed with the 1000 μM catechol solution, while the sides were flat and smooth with the 10 μM solution. As the catechol concentration increased from 10 μM to

1000 μM , the water saturation percentage at the fingertip increased from approximately 70%, to 80%, to 90% and finally to full saturation. Mostly two fingers were formed among the four concentrations of catechol exudate solution. In terms of velocity of the main finger development, the finger velocity increased from 1.63 cm/min at concentration of 10 μM to 2.17 cm/min at concentration of 1000 μM in sequence. Regarding the NaCl control solution (Fig. 9(e)), three fingers were formed, which were mostly

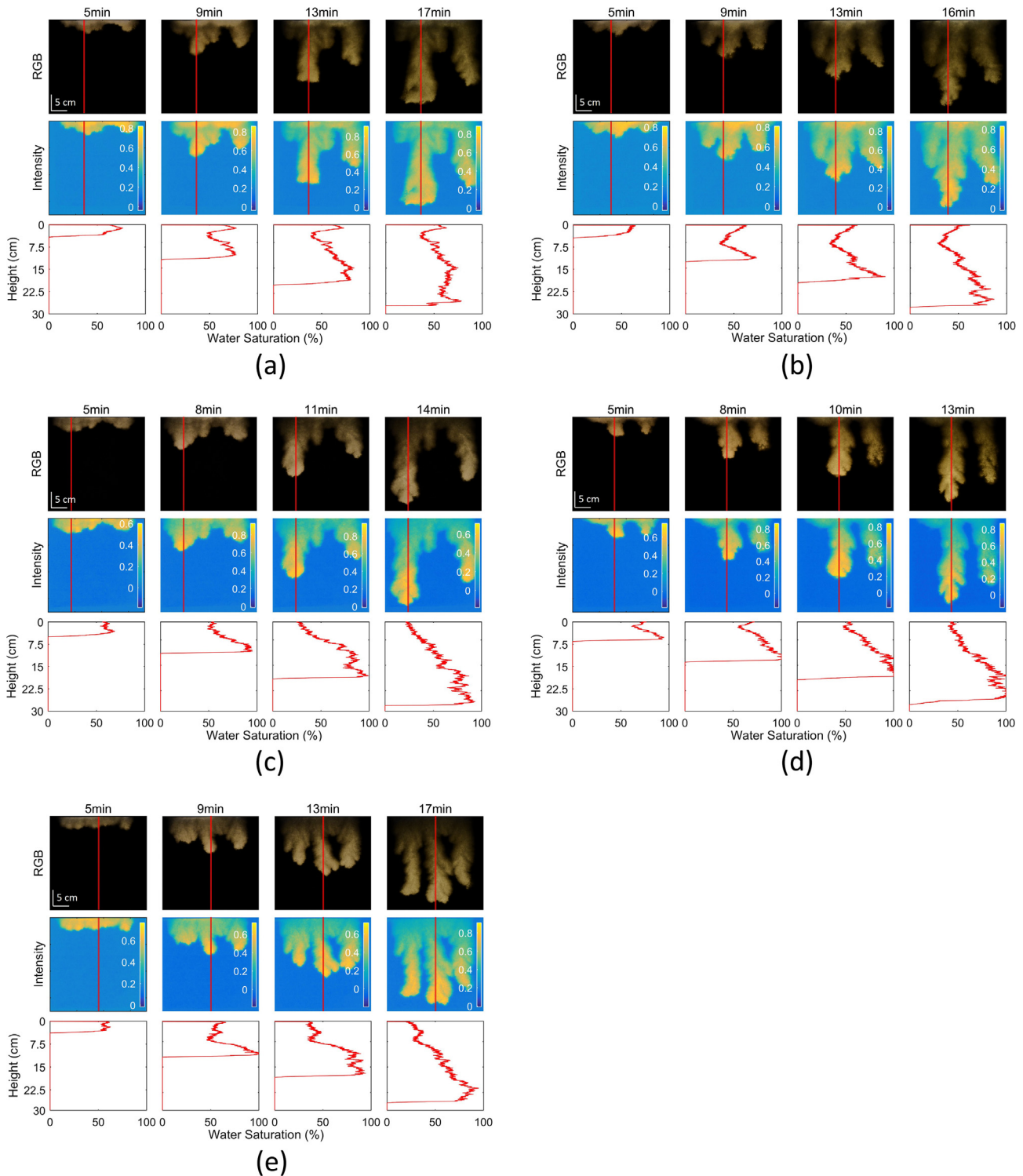


Fig. 9. Vertical water saturation profile analysis for catechol solution at different stages of the main finger formation. Row 1 is the original RGB picture, Row 2 is the Matlab processed image with intensity only, Row 3 is the converted water saturation percentage distribution along the center of the main finger. (a) Concentration at 10 μM ; (b) Concentration at 100 μM ; (c) Concentration at 500 μM ; (d) Concentration at 1000 μM ; (e) NaCl control solution for comparison.

smooth at two sides, and the water saturation percentage at the fingertip was in a similar range with the 500 μM of catechol solution, roughly around 90%.

The horizontal finger profiles of the final stages for four concentrations of catechol as well as the control NaCl solution are collectively shown in Fig. 10. The main finger width at the fingertip was widest for 10 μM catechol solution, at 11.72 cm, and it decreased to a range of five to seven centimeters at higher concentrations. As for the control NaCl solution, the width was 8.4 cm at the fingertip. At 90% of the total length of the main fingers, the water saturation percentage profiles suddenly increased and dropped at both sides where the solution failed to reach the sand matrix. The middle portion of the water saturation percentage profiles increased from approximately 70% to 80%, to 90%, and to almost 100% as the catechol concentrations were raised from 10 μM to 1000 μM , which was consistent with the vertical profiles.

3.2.2. Riboflavin

Vertical and horizontal finger profiles were analyzed for different concentrations of riboflavin solutions at different stages of the main finger formation, beginning at 5 min and terminating when the main finger almost contacted the tank bottom. The vertical profiles by riboflavin concentration for the four successive stages of finger formation, including their original RGB pictures, the corresponding intensity figures and the water saturation percentage distribution profiles, as well as the NaCl control for purposes of comparison, are shown in Fig. 11. An analysis of the main finger shapes indicated a more bulbous main finger within an increase in riboflavin concentration from 10 μM to 500 μM . Further, the water saturation percentage also increased at the fingertip of the riboflavin solutions from 75% to 85% and to 90% as the concentrations increased from 10 μM to 500 μM , with the exception of the 1000 μM solution (which decreased to 80%). Each of the four riboflavin concentrations also formed two fingers during which the velocity was the highest (2.07 cm/min) at 10 μM concentrations and decreased to 1.63 cm/min at 1000 μM concentrations. For a

1000 μM concentration of riboflavin solution, the slightly inclined main finger was relatively thinner compared to the fingers at lower concentrations, and it did not exhibit the highest water saturation percentage as did the fingertip in the last stage. The highest percentage of water saturation in the NaCl control was approximately 90%, which was quite close to the 500 μM of the riboflavin solution.

The horizontal finger profiles for the final stages of the four concentrations of riboflavin and the control NaCl solutions are shown in Fig. 12. The widest main finger width was 9.84 cm at 500 μM , with the width decreasing to 6.97 cm at 100 μM and to 6.66 cm at 10 μM . An approximate 75% water saturation was observed in the horizontal profiles for the 10 μM solution, and it increased to nearly 90% for the 500 μM solution and then decreased to 80% for the 1000 μM solution, which were consistent with the vertical profiles.

4. Discussion

Microbial exudate, as the first step in the simulation of a microbial effect, does affect the infiltration process, particularly preferential flow through porous media. In this research, catechol and riboflavin in particular were studied, as were the interfacial characteristics of contact angle and surface tension which played an important role in the analysis of the fingering flow process.

The study of interfacial characteristics of catechol and riboflavin provided the basic information for their linkage to preferential flow features. The opposite behaviors in contact angle, in addition to different performances in surface tension, induced various flow patterns for two exudate solutions among four concentrations at 10, 100, 500, and 1000 μM (Figs. 7 and 8). The light transmission method was used to correlate the transmitted light intensity to the water saturation percentage, which showed significant differences among the finger profiles (Figs. 9–12 and S1–S4).

In a homogeneous porous medium, preferential flow may lead to a finger-shaped wetted zone filled with water and solutes. The fully formed fingers generated by gravity-driven flow behave like

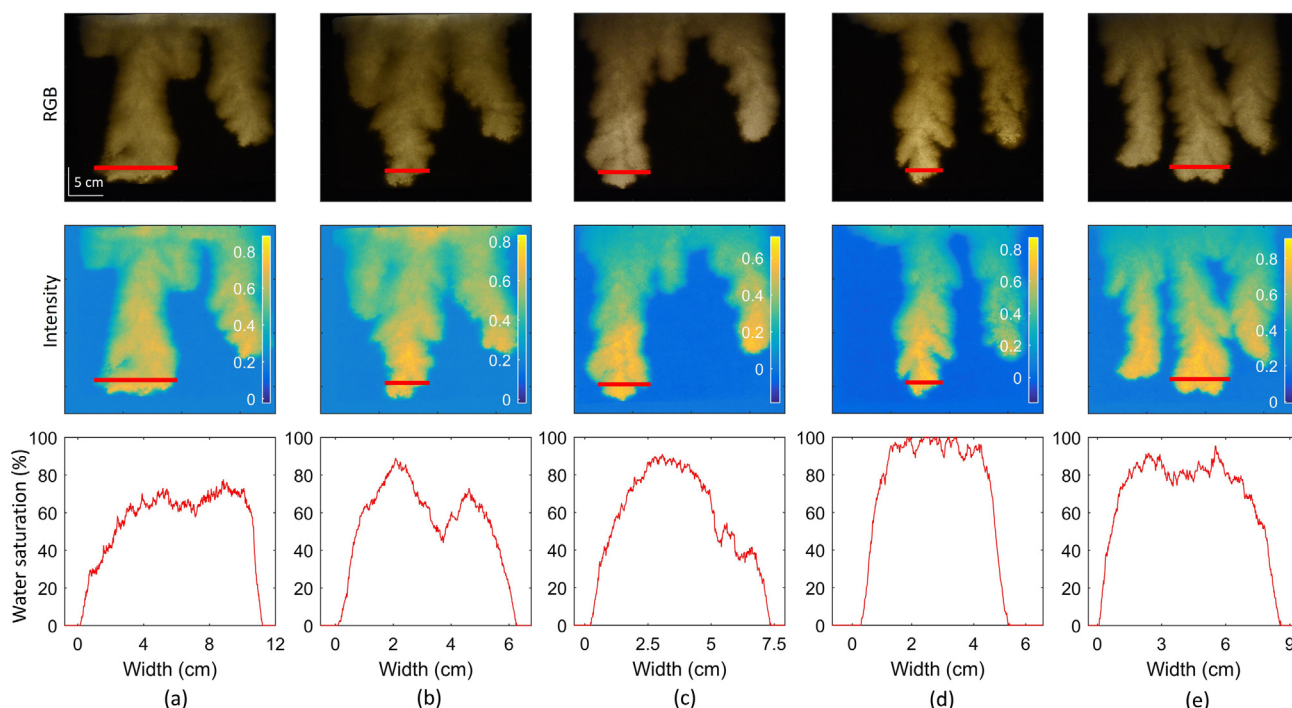


Fig. 10. Horizontal water saturation profile analysis for catechol solution at the final stage of the main finger formation: (a) Concentration at 10 μM ; (b) Concentration at 100 μM ; (c) Concentration at 500 μM ; (d) Concentration at 1000 μM ; (e) NaCl control solution for comparison.

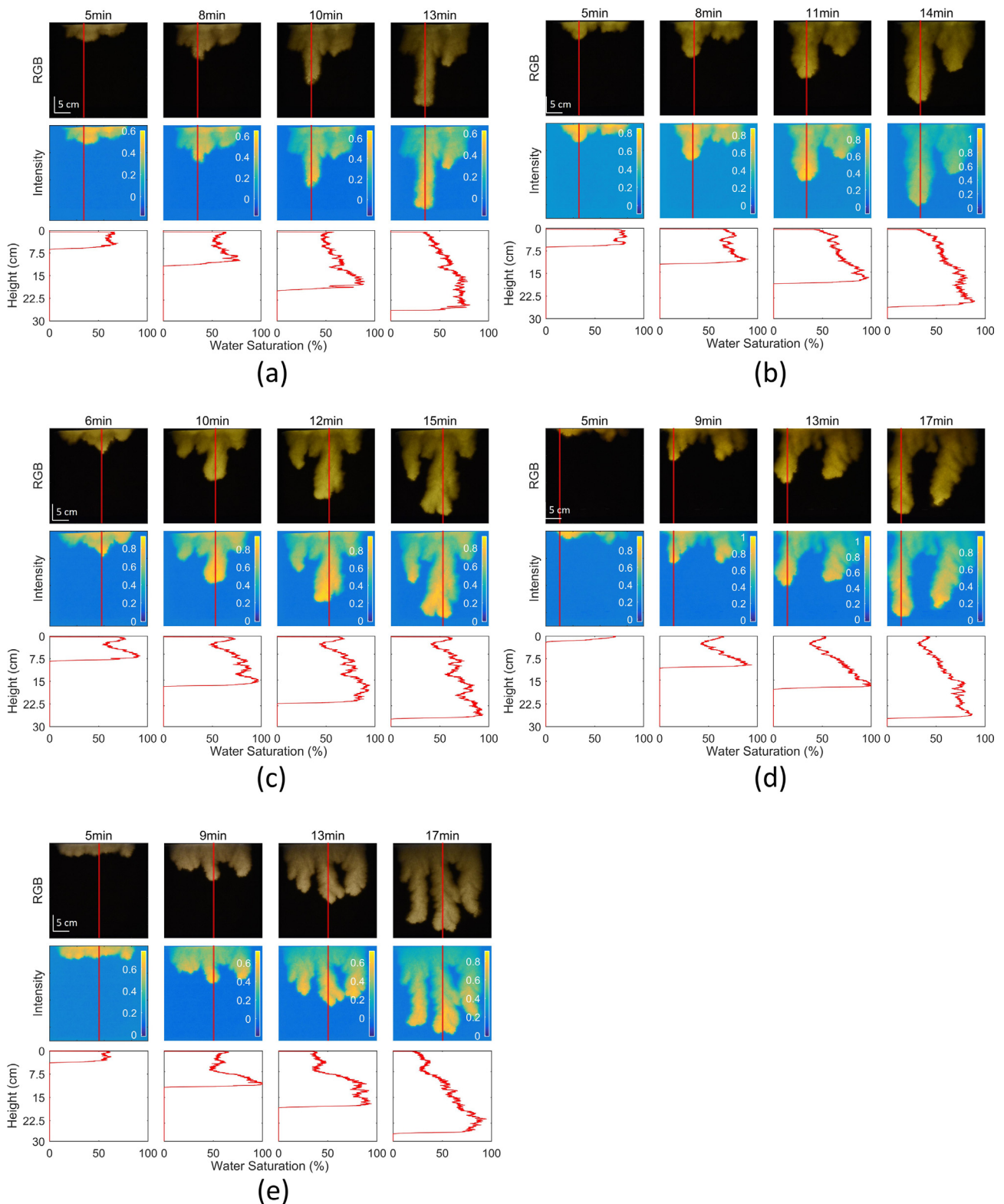


Fig. 11. Vertical water saturation profile analysis for riboflavin solution at different stages of the main finger formation. Row 1 is the original RGB picture, Row 2 is the Matlab processed image with intensity only, Row 3 is the converted water saturation percentage distribution along the center of the finger. (a) Concentration at 10 μM ; (b) Concentration at 100 μM ; (c) Concentration at 500 μM ; (d) Concentration at 1000 μM ; (e) NaCl control solution for comparison.

traveling waves (i.e., constant shape and velocity) and the water saturation profile of the fully formed fingers exhibits the following pattern: a region of high water saturation directly behind the wetting front, known as the fingertip—i.e., an overshoot at the fingertip; followed by another region of drainage where the water

content in the finger decreases to a lower and uniform water saturation, known as the finger tail (DiCarlo, 2004; Glass et al., 1989a; Liu et al., 1994; Selker et al., 1992). In this study, the fingers in the initially dry medium formed wet cores with relatively high water content in contrast to their dry surroundings as initially described

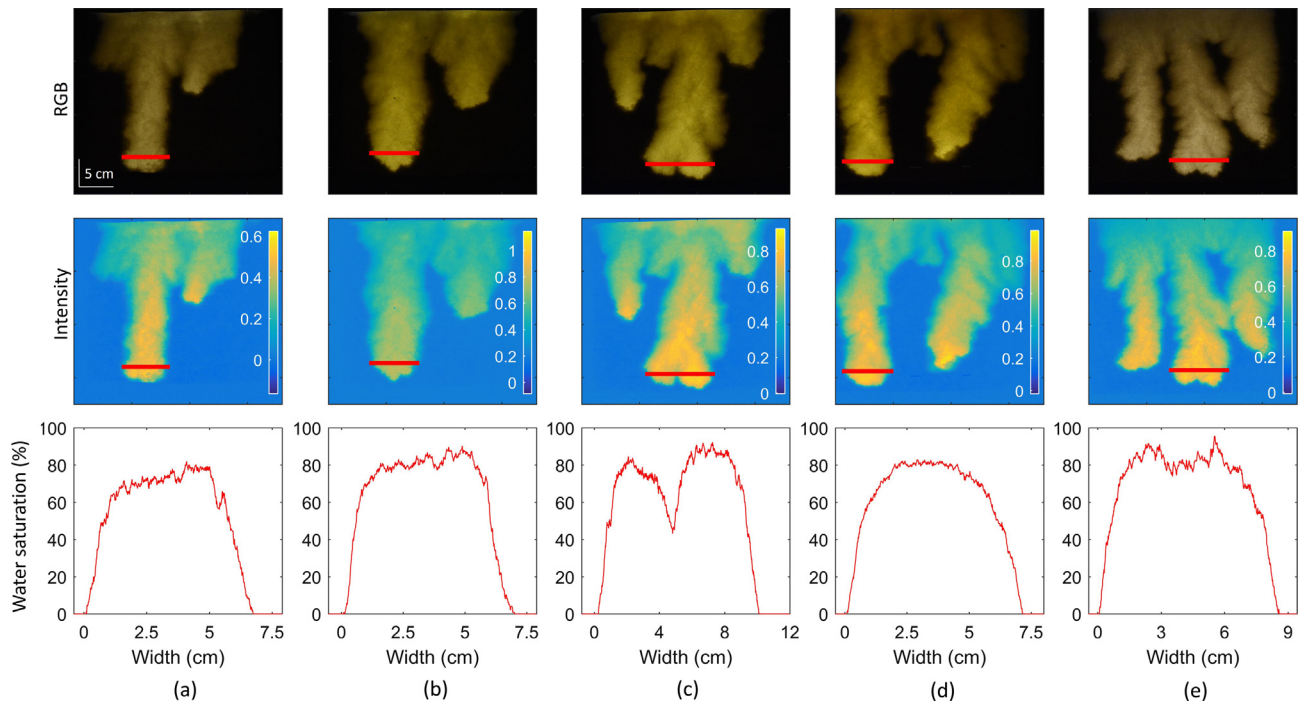


Fig. 12. Horizontal water saturation profile analysis for riboflavin solution at the final stage of the main finger formation: (a) Concentration at 10 μM ; (b) Concentration at 100 μM ; (c) Concentration at 500 μM ; (d) Concentration at 1000 μM ; (e) NaCl control solution for comparison.

by Hill and Parlange (1972). The movement of the main fingers downward, as indicated in the images, failed to disturb the widths of the finger cores for a short distance immediately behind the fingertip (Figs. 10 and 12, and S2 and S4). This stasis was due to the drying process behind the fingertip and reduction of matric potential preventing a rapid expansion of the fingers (Glass et al., 1989c). Similarly, an analysis of the relationship between the width and velocity of the finger determined that a higher flow rate led to a wider and faster moving finger (Glass et al., 1989a,b). In this examination of the flow experiments, under similar infiltration rates, the physical properties of the microbial exudate solutions were deemed the major cause of variability in the preferential flow paths (i.e., fingered flow). The size and shape of the fingers and the water distribution profiles for all the 2D tank experiments provided direct evidence of the influence of the catechol and riboflavin solutions that were analyzed (Table 4, Figs. 9–12 and S1–S4). Each of the four concentrations of respective microbial exudate solution indicated distinct contact angles and surface tensions (Table 3), all of which placed different stresses upon the infiltration process and finger formation.

Wetting front instability has been studied under several scenarios, most notably that of Parlange and Hill (1976) who derived modeling equations to predict the width of a finger in an air-water system by incorporating the sorptivity, and Glass and Nicholl (1996) who adjusted the equation to fit an air-water system. This system is expressed as:

$$d_c = \pi \frac{S^2}{2K_s(\theta_s - \theta_i)} \left(\frac{1}{1 - q_s/K_s} \right) \quad (2)$$

where d_c is the minimum finger width, S is the sorptivity, θ_s is the saturated water content, θ_i is the initial water content, q_s is the flux, K_s is the saturated hydraulic conductivity, and π is a constant for a 2D system and 4.8 for a 3D system (Glass et al., 1991; Wallach et al., 2013). It is important to note that the intrinsic sorptivity is independent of the liquid properties (Philip, 1969; Wallach et al., 2013). Scaling of fluid flow in porous media was developed by Miller and

Miller (1956). Hydrodynamic scaling in porous media applied to fingered flow may be used to characterize the finger dimension. The theoretical analysis of wetting front instability in soils, developed by Parlange and Hill (1976), led to equations that predict the finger dimension (i.e., width) in porous media. These equations of finger dimension can be scaled using the Miller and Miller scaling theory (1956) to obtain a scaling equation that can be applied to width of fingers, expressed as:

$$d_* = \frac{\rho g \lambda}{\sigma_{gl}} d \quad (3)$$

where ρ is liquid density, g is gravitational constant, λ is the characteristic media length scale, σ_{gl} is the gas-liquid interfacial tension, d is the finger width, and d_* is the scaled finger width (Glass and Nicholl, 1996). This scaling equation was previously applied to finger dimensions by varying the characteristic media length scale, the liquid density, and the gas-liquid interfacial tension (Selker and Schroth, 1998). Using this scaling equation, we compared the finger dimensions obtained in our experiment by performing a scaling based on the density of the microbial exudate solutions relative to the control NaCl solution, the gas-liquid interfacial tension of the microbial exudate solution relative to the control NaCl solution, and the coupling of both the density and gas-liquid interfacial tension of the microbial exudate solution relative to the control NaCl solution (Table 5). Scaling results show that the fluid density, the gas-liquid interfacial tension, and the coupling of fluid density and gas-liquid interfacial tension did not have a significant influence on the scaling inconsistencies of the finger dimension (Table 5). As stated by Selker and Schroth (1998), the cause is related to the consistency of the contact angle among the experiments to be scaled when using the Miller and Miller scaling theory. Culligan et al. (2005) showed that soil sorptivity is a function of $(\cos \theta)^{1/2}$. Recently, an extension of the Miller and Miller scaling theory to scale sorptivity by contact angle was proposed by Wallach and Wang (2013). Their results showed that the Miller and Miller equation can be employed for scaling sorptivity by the

Table 5

Hydrodynamic scaling of finger width measurements resulting from microbial exudates infiltration in ASTM graded sand C778. Finger width is scaled to density, gas-liquid interface tension, density and gas-liquid interface tension of pore water, and $(\cos \theta)^{1/2}$.

Solution and concentration	Solution density (g·cm ⁻³)	Porous media (Graded sand from US silica sand)	Pore water velocity (cm·min ⁻¹)	Average measured finger diameter (cm)	Scaling of average measured finger diameter with the solution density relative to pore water (cm)	Scaling of average measured finger diameter with the solution gas-liquid interface tension relative to pore water (cm)	Scaling of average measured finger diameter with the solution density and gas-liquid interface tension relative to pore water (cm)	Scaling of average measured finger diameter with the solution $(\cos \theta)^{1/2}$ relative to pore water (cm)
Control NaCl	0.1 M	0.9914 ± 0.0012	ASTM C778	0.22	7.24	7.24	7.24	7.24
Catechol	10 μM	0.9925 ± 0.0008	ASTM C778	0.22	9.69	9.68	9.74	9.73
	100 μM	0.9934 ± 0.0009	ASTM C778	0.22	7.02	7.01	7.07	7.06
	500 μM	0.9946 ± 0.0007	ASTM C778	0.22	7.41	7.39	7.38	7.36
	1000 μM	0.9950 ± 0.0008	ASTM C778	0.22	5.12	5.10	5.12	5.10
Riboflavin	10 μM	0.9922 ± 0.0015	ASTM C778	0.22	6.96	6.95	7.05	7.04
	100 μM	0.9929 ± 0.0007	ASTM C778	0.22	7.34	7.33	7.41	7.40
	500 μM	0.9932 ± 0.0015	ASTM C778	0.22	7.36	7.35	7.41	7.40
	1000 μM	0.9935 ± 0.0008	ASTM C778	0.22	7.75	7.73	7.79	7.77
Average					7.32	7.31	7.36	7.34
C.V.					1.17	1.17	1.18	1.18

contact angle, i.e. $(\cos \theta)^{1/2}$ for uniform pore geometry. This is the case here, as our experiments are conducted in homogeneous and uniform sandy porous media in which fingered flow occurs. Based on these findings, the finger width was scaled with $(\cos \theta)^{1/2}$ to account for the synergistic effect between contact and pore geometry (Table 5) and expressed as:

$$d_* = d(\cos \theta)^{1/2} \quad (4)$$

Scaling results show that the use of $(\cos \theta)^{1/2}$ decreases the scaling inconsistencies of the finger dimension (Table 5), therefore demonstrating the key contribution of contact angle through the expression of $(\cos \theta)^{1/2}$ in the development and growth of fingers in porous media, the width of fingers, as well as the pore scale consideration in unstable gravity driven finger flow (Steenhuis et al., 2013).

The number of observed fingers (two) was similar for each chemical solution (Table 4). From top to bottom, mostly two fingers were formed in the system with the vertical water saturation profiles for the main fingers exhibiting similar patterns (Figs. 9–12 and S1–S4). From top to bottom of the 2D tank, these main fingers often encompassed two zones, both of which differed from the theoretical finger profile described by Glass et al. (1989c). An ideal finger has three zones of classification indicating a downward migration of the solution in an initially dry medium. The conceptual profile transitioned from a zone with a constant degree of water saturation between 50% and 60% to that of an increasing gradient in water saturation, and which reached near or full saturation at the fingertip. Further, although only the transition and saturation zones were clearly observed in the vertical water saturation profiles of fingers for catechol and riboflavin solutions, all three zones were present in the NaCl control solution (Figs. 9–12 and S1–S4). The difference between the control solution and the exudate added solutions indicates a possible increase of the degree of water saturation in porous medium by the microbial exudate prior to saturation. It is now known that microbial excreted extracellular polymeric substance (EPS) can increase the holding capacity of soil water (Morales et al., 2010), which is consistent with the findings detailed here. Specifically, an application of EPS on soil particles showed an increase in soil-water contact angle and a great change in soil matric potential, which caused the formation of preferential flows (Bauters et al., 2000). Similarly, the use of

commercially prepared dextran and xanthan to model the microbial exudates of polysaccharides in order to analyze the soil structure and water transport indicated a reduction in water repellency of soils after the microbial exudate treatment (Czarnes et al., 2000). Clearly, these studies validated the effect of the microbial exudates on the infiltration process that was detailed in this study.

An analysis of the effect of concentration differences for each microbial exudate solution was also undertaken during the infiltration experiment. It is well known that the underground migration of contaminants is closely related to the properties and chemistries of the solution, meaning that the solute concentration, ionic strength and pH affect both the surface tension and contact angle (Bashir et al., 2009, 2008; Henry and Smith, 2002; Lord et al., 1997). Therefore, the Young-Laplace equation can be used to express capillary pressure at the pore scale as a function of surface tension and contact angle. This equation is expressed as:

$$P_c = \frac{2\gamma \cos \theta}{r} \quad (5)$$

where P_c is for the capillary pressure, γ and θ are the surface tension and contact angle of a solution, respectively, and r is the radius of the pore size of medium (Lord et al., 1997).

To explain the action of the interfacial forces upon the solution's infiltration process, we assumed a constant radius for pore size under various combinations of surface tension and contact angle. The capillary pressures would then correspondingly contribute to water flow and distribution changes across the porous medium, resulting finally in diverse infiltration processes and finger shapes. Here, a decrease in contact angle at higher concentrations of catechol solution (Table 3) likely increased the wettability of the porous medium thus causing a greater protrusion of the solution into the dry matrix and quickly reaching conditions of near saturation. The slopes of the transition zone shown in the vertical water saturation profiles increase with increasing catechol concentrations which reflect this faster water distribution rate (Fig. 9). Based on these horizontal water saturation profiles for the catechol solutions under study, the water saturation distribution profile at 90% of the total finger length of the highest catechol concentration of 1000 μM showed the greatest drops at two edges, the greatest constancy of degree of water saturation in between, and the narrowest width (Fig. 10). As expressed in Eq. 5 at 1000 μM of catechol solution for example, the largest cosine of the contact angle, which was

given by the smallest contact angle value, multiplied by the second small surface tension nearly cancelled out its resulted capillary pressure change (Figs. 3 and 4). Therefore, the capillary pressure may not be the causative factor in the finger development process among concentrations for catechol. A better water holding capacity induced by a catechol solution at higher concentrations may be useful in explaining the greater wettability and water saturation percentage at fingertip, and a reduction in drainage to the surrounding dry matrix (Fig. 10). Here, the water holding capacity resulting from catechol may dominate the preferential flow patterns over the interactive forces that occur at the interfaces of soil-water solution and sand grain, and of soil-water solution and air.

An increase in contact angle that was observed with an increase in concentration of riboflavin (Table 3) indicated that more time was required within the medium to form a fully developed main finger due to decreased wettability. The difference in the slope of the transition zone, the water saturation at the fingertip (Fig. 11), and the horizontal water distribution profile between four riboflavin concentrations was not as apparent as in the catechol solutions (Fig. 12). However, a slight increase in saturation percentage at the fingertip and in the overall finger width shown in riboflavin's vertical and horizontal water saturation profiles from 10 μM to 500 μM could still be observed, with an exception of 1000 μM where the finger might have been affected by the incline. While the surface tension from the catechol solutions indicated no obvious trend among concentrations, it decreased when the riboflavin concentration increased from 10 μM to 1000 μM (Fig. 6). In the context of Equation 5, at 10 μM of riboflavin solution, for example, the smallest contact angle, which led to the largest value of its cosine, in addition to the largest surface tension resulted in the largest capillary pressure value (Figs. 5 and 6). At higher concentrations for riboflavin, this reducing capillarity pressure therefore enhanced water saturation except for 1000 μM . A wider finger width at higher concentration except for 1000 μM may be due to a decrease in water holding capacity, with more water diffused into the dry soil matrix. Moreover, by adding chemicals, the surface tension of water could decrease and therefore the fingers' size would reduce (Glass et al., 1989a). Although the phenomenon described by Glass et al. (1988) is opposite to our findings, the study above further supports the likelihood that surface tension also plays a role in changing the preferential flow process.

Previous point source experiments have determined that at the same flow rate with a lower contact angle of media, the fingers tend to be more bulbous and wider than under a higher contact angle (Wallach et al., 2013). However, our observance of the fingers from the catechol and riboflavin solutions (Figs. 9 and 11) indicated a decreased contact angle that narrowed the bulbous finger shapes with an increase in catechol concentration and a decrease in riboflavin concentration (Table 3). The results presented here may differ from previous work due to the imposed boundary condition of simulated rain event induced infiltration rather than point source dripping infiltration. The application of the random rain event completely saturated the top layer. Consequently, the capillary force acting on the wetting front that caused the finger to spread was slightly smaller as compared to the gravitational force. Therefore, for the lower contact angles, the bulbous shape failed to persevere, as gravity caused the eventual elongation of all fingers to different degrees at the respective contact angles and surface tensions.

As fingers flow through a porous medium, the water carries solutes and nutrients to microbes in pores and, in turn, the microbes and their exudates possibly affect the finger flow pattern and water distribution. As a degraded intermediate of aromatic compound, catechol is poisonous and possibly harmful to microbes. As the catechol concentration increases from 10 μM to 1000 μM , the fingers become narrower and produce more

branches (Fig. 10). A possible relationship between microbial behavior and fingering flow may manifest in the microbes metabolizing to more rapidly dispense with the catechol, as they increase the contact area to the surrounding matrix and minimize or narrow the internal hold zone for the catechol. An increased water saturation percentage at higher concentration may also be useful in catalyzing the chemical reactions in a wet system. However, riboflavin also serves as a nutrient for microbes. Specifically, in studies with riboflavin at increased concentrations from 10 μM to 1000 μM , the fingers were observed to grow wider except for the 1000 μM sample (Fig. 12). A possible explanation is that the riboflavin exudate effect on the preferential flow is tied to the degree to which the microbes favor this nutrient. At higher concentrations of riboflavin solution (500 μM), microbes might favor this nutrient and wish to retain it for further use, which leads to wider fingers and a maximum internal zone. However, once the concentration reaches 1000 μM , which might be too high for microbes, the flow pattern becomes narrow again. The highest water saturation percentage at 500 μM of concentration observed in profiles created the wettest system for the maximum microbial activity.

Thus far, this research used the light transmission method to evaluate the concentration effects of artificial microbial exudates on infiltration process, which has not been widely studied. The use of the light transmission method, an optical imagery system using visible light, has been used in such diverse applications as contaminant hydrogeology, measurement of flow paths, and in determining the velocity of both fluid and solutes, and in characterizing both fluid distribution and contaminant transport (Werth et al., 2010). Boundary effects on the hydrodynamics process observed in the two-dimensional tank may arise from the contact of flow and porous media with the plates. These boundary effects are minimized by choosing a sand grain size that is relatively small compared to the width of the two-dimensional tank to ensure that sufficient porous media size between the two plates is available for the development of the flow in porous media. Many such optical methods have been used to evaluate many biotic and abiotic systems in either saturated or unsaturated porous media. However, very little research has been undertaken on unsaturated systems, and only a few studies emphasizing the process of infiltration (Keller and Auset, 2007). Of the studies analyzing biomass growth, microbial and colloidal transport, and biofilm process in unsaturated porous media, most involved only optical methods to qualitatively visualize pore-scale flow or transport, with additional results derived from a further model calculation or graphing breakthrough curves (Auset et al., 2005; Dunsmore et al., 2004; Sirivithayapakorn and Keller, 2003). The application of the light transmission method in this research directly related image information (i.e., light intensity) to flow characteristics (i.e., water saturation percentage) quantitatively through calibration. The study also filled the gap and expanded the research regarding infiltration process related to microbial activity, where the previously mentioned study looked solely at the contact angle influence (Wallach et al., 2013). This paper, however, was more specific regarding types and concentrations of fluid. Infiltration, as one of the most important processes that occurs in natural systems, is worth investigating, given its effects on both biotic and abiotic characteristics in the subsurface. Future research must entail the use of actual microbes and different biomass concentrations which closely corresponds to the various microbial exudate concentrations. A departure from the analysis of microbial transport, biofilm growth, or bioclogging would involve studying the effect of microbial existence on the infiltration process in nearby flowing water.

Furthermore, the evolution of the research involving microbes should emphasize the secreted microbial exudates diffusing from those microbes, which may cause a change in the water gradient

potential around the microbial colony. A study using the light transmission method to reveal the microbial exudates' distribution around living microbes and to evaluate the relationship between media moisture and exudates' diffusion process (e.g., rate) would be quite interesting. The integration of hysteresis with repeated wetting and drying cycles to investigate the influence of microbes and microbial exudates is also recommended. The image and data acquired from imaging techniques will be useful for depicting colony morphology in porous media, to show the shifting variations in water content between dry and wet conditions, and to explain the water uptake strategies used by microbes under changing media hydrological properties (Carminati et al., 2016). As new bioimaging technology of microbial systems are developed, these technologies may be coupled with the light transmission method to study multiscale dynamics of water saturation and transport regulated by microbes. The light transmission method may also be used to support research on microbial competition and evolution in natural porous media, as this research endeavor "will require the development of more realistic models, and additional data associated with the physical, chemical, and microbial characteristics of microenvironments in real porous media" (Baveye and Darnault, 2017).

5. Conclusion

The important findings of this 2D tank microbial exudate flow study are summarized below:

- The light transmission method is a unique tool allowing the investigation of the synergistic influence of solution chemistry and porous media characteristics on the infiltration process, including the unstable fingered flow phenomenon, with a high spatial and temporal resolution.
- Microbial exudates affected the preferential flow during the infiltration process by forming two zones in the vertical water saturation profiles for both of the artificial exudate-added solutions, catechol and riboflavin, rather than three zones in the NaCl control solution. This formation occurs perhaps from the enhancement in the water retaining capacity of the porous medium.
- An examination of the infiltration process under the same intake flow rate determined the influence of various concentrations upon the preferential flow, which was indicated by different vertical and horizontal profiles of the degree of water saturation within the fingers.
- The lowest catechol solution concentration (10 μM) produced the largest finger width (9.69 cm) among the tested catechol solution concentrations, while the highest riboflavin solution concentration (1000 μM) generated the highest finger width (7.75 cm) among the tested riboflavin solution concentrations.
- The results of this study demonstrated the influence of the contact angle expressed as $(\cos \theta)^{1/2}$ on the application of the theory for hydrodynamic scaling in homogeneous and uniform porous media and applied to the scaling of the finger dimension, i.e., finger width.
- The interfacial measurements and distinguished flow patterns of the two microbial exudates indicated each had a different reaction upon the infiltration process.
 - o An increased wettability of the catechol solutions was caused by a reduced contact angle at higher concentrations, creating more branches in the fingers that greatly accelerated the time to saturation. A faster finger moving velocity, a greater water saturation percentage, and a narrower finger width were observed at higher concentrations. An increase in water holding capacity at higher concentration was hypothesized as the main cause of this phenomenon.

- o The contact angle within the riboflavin solutions increased with a corresponding increase in concentration, decreasing the wettability at higher concentrations. A slower finger moving velocity, a small increase in water saturation percentage at the fingertip, and a larger width of the overall finger size was observed as the riboflavin concentration increased from 10 μM to 500 μM . The reduced capillarity gave a greater saturation at fingertip at higher concentrations of riboflavin.
- Possible relationships between the microbial activity and preferential flow characteristics were assumed according to the properties of chemicals.
 - o Microbes may try to minimize contact with catechol and tend to affect the preferential flow by narrowing the finger.
 - o Microbes favor the usage of riboflavin and tend to increase the width of fingers.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.jhydrol.2017.12.065>.

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